

Reference List Entry:

Sharma, S. and Panda, S.P. (2023) 'Efficient information retrieval model: overcoming challenges in search engines-an overview', *Indonesian Journal of Electrical Engineering and Computer Science*, 32(2), pp. 925–932. doi:10.11591/ijeecs.v32.i2.pp925-932.

Svore, K.M. and Burges, C.J.C. (2009) 'A machine learning approach for improved BM25 retrieval', in *Proceedings of the 18th ACM Conference on Information and Knowledge Management (CIKM '09)*. Hong Kong, China, 2–6 November 2009. New York: ACM, pp. 1811–1814. doi:10.1145/1645953.1646237.

Kasenchak, R.T. (2019) 'What is Semantic Search? And why is it important?', *Information Services & Use*, 39(3), pp. 205–213. doi:10.3233/ISU-190045

The Models

As part of my SS2 problem statement for Stadioalot I have investigated and developed two information retrieval models. Namely BM25 and Semantic Search Model

According to Sharma and Panda (2023) Information Retrieval Models are a theoretical , mathematical or algorithmic framework. That specifies how search systems represent documents and user queries match them based on semantic or statistical representation and rank the information that was retrieved .

According to the journal article Information Retrieval is the backend of how search engines , databases and digital libraries operate and find relevant information . The principles of these models is to index raw data to a structured searchable format than matches the relevance of user queries and lastly ranks and scores the documents from highest to lowest (Sharma and Panda, 2023)

For the case and the demonstration I used BM25 which is a lexical search model. So what is a Lexical Search Model? According to Svore and Burges (2009) Lexical Search is a traditional retrieval technique that uses exact words or text overlap between search queries and indexed documents . This point of the BM25 being a more traditional model I think is demonstrated in Part A of the product where 2/3 articles studied used this model as a baseline model. BM25 (Best Matching 25) According to Svore and Burges BM25 is a Fast , precise model that excels in exact domain specific terminology however struggles with human language nuances. This was a point that was made in my SS1 submission of the nuances of South African languages and the examples of if a user were to search for Takkies and not Running shoes and a Seller would list the item in their product description.

The second model was Sementic Search however is an advance information retrieval model that understands intent , context and the meaning of a search query rather than the literal keywords (Kasenchak, 2019) Kasenchak notes semantic searches shift from literal text strings towards understanding real world concepts , entities and relationships by understanding natural language. Sentence Transformer Embedded Models are deep learning neural networks that map queries into dense mathematical vectors (embedding). Sentence Transformers do not convert single words into numbers but rather evaluate entire sentences and are able to capture semantic nuance and context.

The datasets

From the WANDS TSV file , we have three datasets

- 1.Product df , which is the product catalogue of what the store has and what can be search for by customer.42994 products
- 2.The second dataset is the query df of what customers have actually searched for 480 queries
- 3.And then the final dataset is the label df of human relevance search results and the key evaluation that our model will be evaluated on 233 448 human labeled query and product relevance pairs with the label being , exact , partial or irrelevant match

Models	BM25	Sementic Search
Data Preparation	Load all three datasets Than asses the quality of the dataset of the raw datasets, what values are missing by inspecting the dataset Fiilna , NaN for any row with no text eg product descriptions for chair has no features : NaN This might also affect the ability of BM25 to create the lexical matching model if the product has missing descriptions Breakdown words (create df into search document All product info was in different doc now we have a new feature called search document all the info in one place and its this info that feeds the model Remove punctuation	Rather than a lexical term match a semantic model understand the users search intent In the prep we install the semantic library , this took about one hour because of how massive the the product catalog is of the WANDS product catalogue In this phase we installed a sentence transformer library that has pre pre-trained sentence transformer that converts text to numerical embedded models . The embedding vector is than represents the meaning and context of text to compare product embedding and query embedding We than imported pandas and the three WANDS datasets , but for this model preparation we do not need to tokenization , instead we load pretrained semantic model that converts 42 994 product descriptions to embeddings

Feature engineering	At this phase all the columns of the product database are removed and the dataset will be one big searchable text as illustrated in my notebook. This is called a search document. And it combines metadata from the dataset and looks at , then we have document length , product length and query length. Then we install the BM25 model itself the main job of the model is tokenization .The tokenization is the first step in processing text before the BM25 model can calculate or rank raw text (in this case the product dataset) into individual token so the model is able to 'count and score' them	Then we load the semantic search model and create embedding for the products so instead of comparing the product to the query we are now looking at the vector representing meaning which all the text have a number and together they represent a semantic of the text and than the query also get converted into a vector and all products that have the same meaning as the query move towards the top For the feature engineering we have Product search document similar to model BM25 model where all the text has been combine from all the columns in the database Secondly the pre trained sentence transformer converted all product searches document into a 384 dimensional embedded vector The 384 dimension comes down to the underline neural network architecture